

#HW 4

● Graded

Student

Rayyan Khan

Total Points

100 / 100 pts

Question 1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts quality not found out.

- 2 pts The final state temperature is not correct.

- 2 pts Quality found out is not correct.

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts The work calculation is wrong.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 5

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts correct answer 71 degree C approx.

- 2 pts solution not shown.

Question 6

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 3 pts incomplete

Question 7

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 5 pts incomplete

Question 8

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts Correct answer:0.95 cubic m

- 4 pts incomplete

Question 9

(no title)

10 / 10 pts

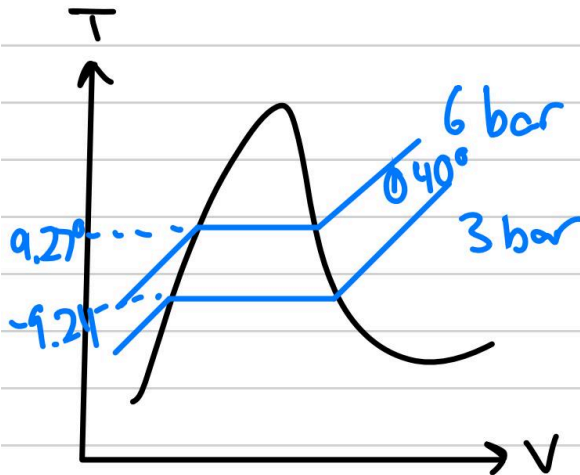
✓ - 0 pts Correct

- 2 pts Correct answer: 10 bar approx.

- 3 pts incomplete

Question assigned to the following page: [1](#)

- (a) Ammonia, initially at 6 bar, 40°C undergoes a constant-specific volume process to a final pressure of 3 bar. At the final state, determine the temperature, in °C, and the quality. Locate each state on a sketch of the $T-v$ diagram.



$$\Delta w = 0, \Delta u = Q$$

	Phase 1	Phase 2
p	6	3
T	40°	
State	Superheated vapor	liquid vapor mix
v	0.24118	0.24118

at 3 bar $v_f = 1.5361 \times 10^{-3} \text{ m}^3/\text{kg}$ and $v_g = 0.4061$

Since 0.24118 m^3 is between those specific volumes, it is a liquid vapor mix

Since it's a liq-vap mix $T_2 = T_{\text{sat},2} \rightarrow T_2 = -9.24^\circ\text{C}$

$$0.24118 = (1-x)(1.5361 \times 10^{-3}) + x(0.4061)$$

$$x = 0.592$$

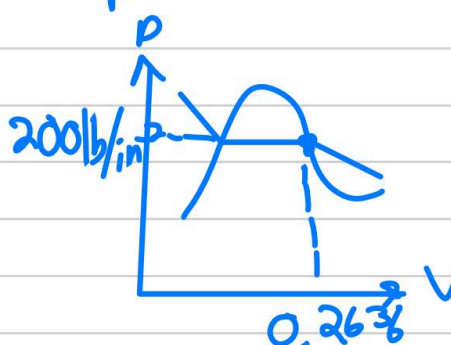
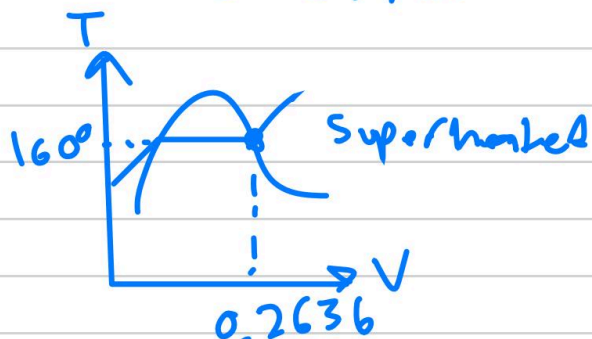
Question assigned to the following page: [2](#)



For each case, determine the specified property value and locate the state sketches of the p - v and T - v diagrams.

- (a) For Refrigerant 134a at $T = 160^\circ\text{F}$, $h = 127.7$ Btu/lb. Find v , in ft^3/lb .
- (b) For Refrigerant 134a at $T = 90^\circ\text{F}$, $u = 72.71$ Btu/lb. Find h , in Btu/lb.
- (c) For ammonia at $T = 160^\circ\text{F}$, $p = 60$ lbf/in.² Find u , in Btu/lb.
- (d) For ammonia at $T = 0^\circ\text{F}$, $p = 35$ lbf/in.² Find u , in Btu/lb.
- (e) For Refrigerant 22 at $p = 350$ lbf/in.², $T = 350^\circ\text{F}$. Find u , in Btu/lb.

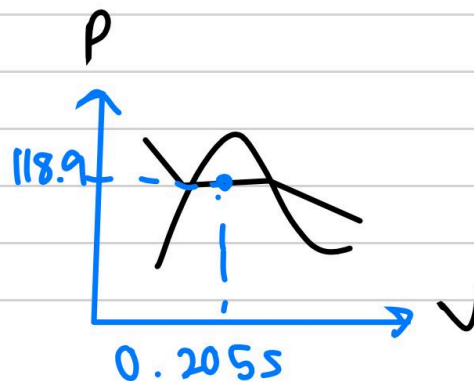
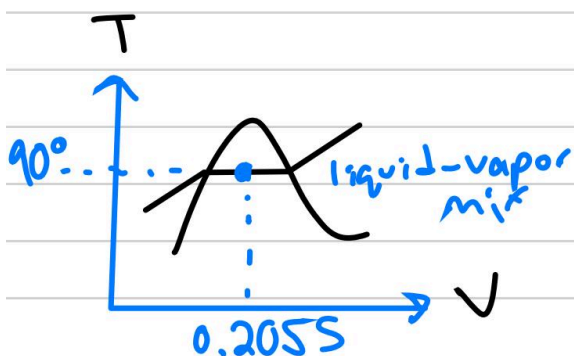
a. From Table A-12, we get $V = 0.263 \text{ ft}^3/\text{lb}$ at $P = 200 \text{ lbf/in}^2$ as superheated vapor.



$$b. \quad x = \frac{u - u_f}{u_g - u_f} \rightarrow \frac{72.71 - 40.42}{105 - 40.42} \rightarrow 0.5$$

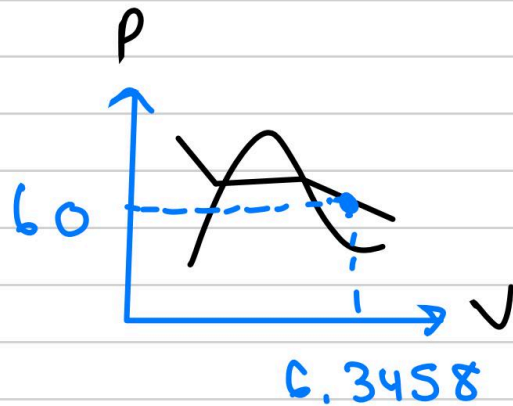
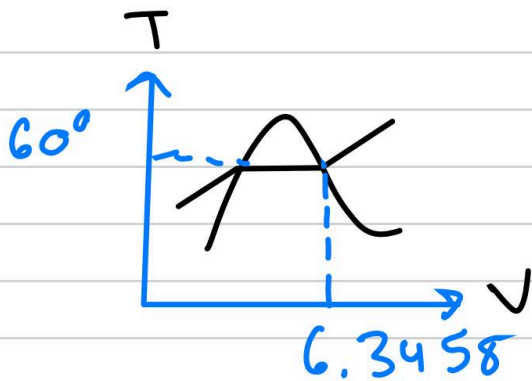
$$h = h_f + x h_g$$

$$= 40.72 + (0.5)(73.03) \rightarrow h = 77.235 \text{ Btu/lb}$$



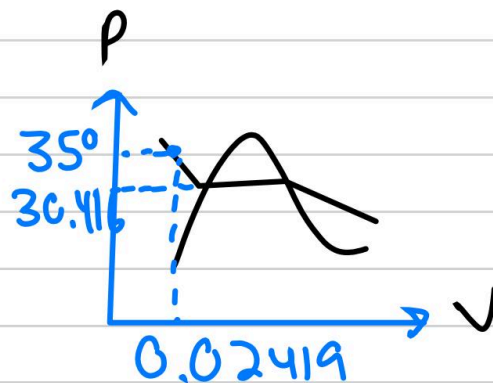
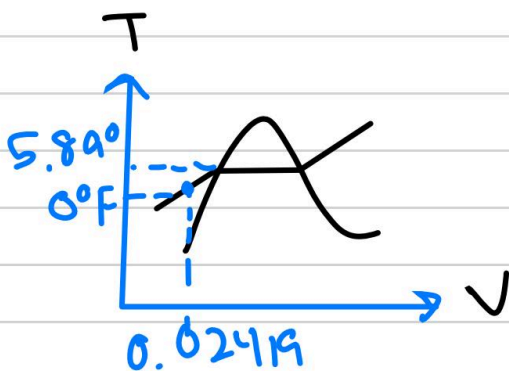
Question assigned to the following page: [2](#)

c. Using Table A-15, $u = 624.44 \text{ Btu/lb}$

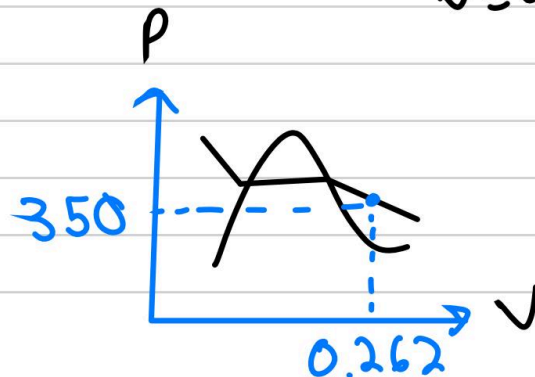
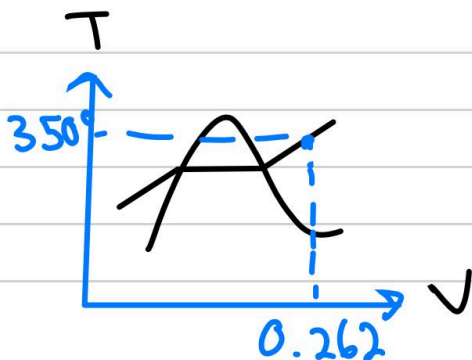


d For compressed, $u(T, p) \approx u_f(T)$, $v(T, p) \approx v_f(T)$

So $u = 42.32 \text{ Btu/lb}$ from A-13

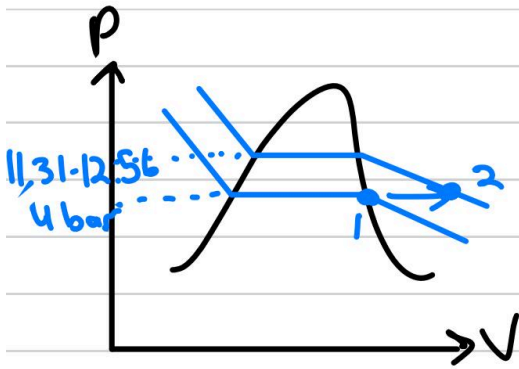


e. By interpolating A-9E, $u = 143.14 \text{ Btu}$ and $v = 0.262 \text{ ft}^3/\text{lb}$



Question assigned to the following page: [3](#)

- (a) Refrigerant 22 undergoes a constant-pressure process within a piston-cylinder assembly from saturated vapor at 4 bar to a final temperature of 30°C. Kinetic and potential energy effects are negligible. For the refrigerant, show the process on a $p-v$ diagram. Evaluate the work and the heat transfer, each in kJ per kg of refrigerant.



Between 28-30°C, $P_{\text{sat}} = 11.313$ to 12.556 bar, which is above the 4 bar at state 2. This means state 2 is superheated vapor. There is constant p from state 1 $\rightarrow$ 2.

State 1: saturated vapor

$$v = v_g = 0.0581 \text{ m}^3/\text{kg}$$

$$u = u_g = 222.24 \text{ kJ/kg}$$

$$4 \text{ bar} = 400 \text{ kPa}$$

State 2: superheated vapor

$$v = 0.06892 \text{ m}^3/\text{kg}$$

$$u = 245.73 \text{ kJ/kg}$$

$$W_{12} = \int_1^2 p dv \Rightarrow p(v_2 - v_1) = 400(0.0682 - 0.0581)$$

$$= 4.248 \text{ kJ/kg}$$

$$Q = W + (u_2 - u_1) \Rightarrow 4.248 + (245.73 - 222.24)$$

$$Q = 27.738 \text{ kJ/kg}$$

Question assigned to the following page: [4](#)

- (a) A closed, rigid tank is filled with 0.02 lb of water, initially at 120°F and a quality of 50%. The water receives 8 Btu by heat transfer. Determine the temperature, in °F, pressure, in lbf/in.², and quality of the water at its final state.

Since rigid tank, $v_1 = v_2$ and $w_{1,2} = 0$

$$v_f = 0.0162 \text{ ft}^3/\text{lb}, v_g = 203.0 \text{ ft}^3/\text{lb}, u_f = 87.99 \text{ Btu}$$

$$u_{fg} = 1049.9 - 87.99 = 961.91 \text{ Btu/lb}$$

$$\begin{aligned} v_1 &= x v_g + v_f (1-x) \\ (1.5)(203.0 + 100.1621)(1-0.5) &= 87.99 + x(961.91) \\ v_1 &= 101.5 \text{ ft}^3/\text{lb} \end{aligned} \quad \left\{ \begin{aligned} u_1 &= u_f + x u_{fg} \\ &= 87.99 + 51901.98 \\ u_1 &= 568.945 \text{ Btu/lbm} \end{aligned} \right.$$

$$Q_{1,2} = m(u_2 - u_1) + w_{1,2}$$

$$8 = 0.02(u_2 - 568.945) + 0 \Rightarrow u_2 = 968.945 \text{ Btu/lbm}$$

From Superheated water table, for $v_2 = 101.5$ at all pressure u is much higher than $u = 968.94 \text{ Btu/lbm}$. This liquid vapor mix.

$$\frac{101.47 - 118.70}{0.000062648} = 0.90 = x$$

$$T_2 = 120.12^\circ \text{F}$$

$$p_2 = \frac{859.07947.22}{11.51 - 8}$$

$$p_2 = 1.694 \text{ lb/in}^2$$

Question assigned to the following page: [5](#)



- (a) At what temperature does water boil on the top of Mount Everest, elevation $z = 8848 \text{ m}$? The dependence of pressure with altitude is:

$$P = P_{\text{atm}} \exp\left(-\frac{MWgz}{RT}\right)$$

where, P_{atm} is atmospheric pressure, g is the gravitational acceleration, and MW is the molecular weight of the gas.

$$MW = 28.97 \text{ g/mol} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 0.02897 \text{ kg/mol}$$

$$\frac{(0.02897)(9.81)(8848)}{T} = (40800)\left(\frac{1}{373.15} - \frac{1}{T}\right)$$

$$= \frac{0.06163}{T}$$

$$= \left(\frac{1}{373.15} - \frac{1}{T}\right)$$

$$T = 350.1524$$

$$T(^{\circ}\text{C}) = 350.152 - 273.15 = 77^{\circ}\text{C}$$

Question assigned to the following page: [6](#)



(a) Water is cooled in a rigid closed container from the critical point to 10 bar. Determine the quality of the final state.

$$P_c = 220.9 \text{ bar}, T_c = 374.14$$

$$\text{Final state: } P = 10 \text{ bar or } 1 \text{ MPa}, T = 179.91^\circ\text{C}$$

Since $T < T_c$, it is a liquid vapor mix

$$\text{When critical, } v_{\text{initial}} = 3.155 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$v_f = 1.1273 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$v_g = 0.1944 \text{ m}^3/\text{kg}$$

Rigid container means $v_1 = v_2$

$$3.155 \times 10^{-3} = x v_g + v_f (1 - x)$$

$$3.155 \times 10^{-3} = x(0.1944) + (1.1273 \times 10^{-3})(1 - x)$$

$$x = 0.01049$$

Question assigned to the following page: [7](#)

Using linear interpolation, estimate the specific volume of water under the following conditions using data from the steam tables. Look up the specific volumes of water that correspond to cases (a), (b), and (c) on the website <http://webbook.nist.gov/chemistry/fluid/> and report their values. Comment on the agreement between the two sources.

(a) $P = 1.9 \text{ MPa}; T = 250^\circ\text{C}$

(b) $P = 1.9 \text{ MPa}; T = 300^\circ\text{C}$

(c) $P = 1.9 \text{ MPa}; T = 270^\circ\text{C}$

a. using table 29.3, I interpolate for $P = 1.9 \text{ MPa}$ which gives me $T_{\text{sat}} = 209.76^\circ$ since $209.76 < 250^\circ$, the water is overheated steam

I then went to table 29.5 super heated water vapor and interpolated for $P = 1.9 \text{ MPa}$ b/c there is only a 1.8 MPa and 2.0 MPa pressure and got

$$\frac{0.11144 - 0.12497}{2 - 8} = \frac{0.11144 - V_{\text{calc}}}{2 - 1.9}$$

$$V_{\text{calc}} = 0.188205 \text{ m}^3/\text{kg}$$

$$V_{\text{read}} = 0.11791 \text{ m}^3/\text{kg}$$

$$\% \text{ error} = \frac{V_{\text{calc}} - V_{\text{read}}}{V_{\text{read}}} \times 100\% = 0.25\%$$

Question assigned to the following page: [Z](#)

b. Same process

$$\frac{0.12547 - 0.14021}{2 - 1.8} = \frac{0.12547 - V_{calc}}{2 - 1.9}$$

$$V_{calc} = 0.13284 \text{ m}^3/\text{kg}$$

$$V_{real} = 0.13249 \text{ m}^3/\text{kg}$$

$$\% \text{ error} = \frac{0.13284 - 0.13249}{0.13249} \times 100\% \rightarrow 0.26\%$$

c. Isobaric, so $T_{sat} = 209.76^\circ$ which is less than 270°C , So it is still overheated

at 1.8 MPa ; $V =$

$$\frac{0.14021 - 0.12497}{300 - 250} = \frac{0.14021 - V}{300 - 270}$$

$$V_{calc} = 0.131066 \text{ m}^3/\text{kg}$$

at 2.0 MPa

$$\frac{0.12547 - 0.11144}{300 - 250} = \frac{0.12547}{300 - 270} \rightarrow V_{calc} = 0.117052$$

→

Question assigned to the following page: [7](#)

Now interpolating for V at 1.9 MPa and 270

$$\frac{0.117052 - 0.131066}{2 - 1.8} = \frac{0.117052 - V}{2 - 1.9}$$

$$V_{\text{calc}} = 0.124059 \text{ m}^3/\text{kg}$$

$$V_{\text{real}} = 0.12389 \text{ m}^3/\text{kg}$$

$$\% \text{ error} = \frac{0.124059 - 0.12389}{0.12389} \times 100\% = 0.1364\%$$

Question assigned to the following page: [8](#)



- (a) A rigid container of volume 1 m^3 contains saturated water at 1 MPa . If the quality is 0.10 , what is the volume occupied by the vapor?

$$V_f = 0.001127 \text{ m}^3/\text{kg} \quad V_g = 0.19444 \text{ m}^3/\text{kg}$$

$$V = V_f + x(V_g - V_f) \\ = 0.001127 + (0.1)(0.19444 - 0.001127)$$

$$V = 0.0204583 \text{ m}^3/\text{kg}$$

$$\text{But } V_{\text{container}} = 1 \text{ m}^3$$

$$m = \frac{1}{0.0204583} \rightarrow 48.88 \text{ kg}$$

$$m_v = x \cdot m \rightarrow (0.1)(48.88) = 4.88 \text{ kg}$$

$$V_v = m_v \cdot V_g \rightarrow 4.88 \text{ kg} \times 0.19444 \text{ m}^3/\text{kg}$$

$$V_v = 0.95 \text{ m}^3$$

Question assigned to the following page: [9](#)



- (a) 40 g of water are sealed in a 10 L container at 300°C. As accurately as you can, determine the pressure of the container.

$$n = \frac{m}{M} \rightarrow \frac{40\text{g}}{18.016\text{g/mol}} \rightarrow 2.22\text{mol}$$

$$R = 0.082057$$

$$\text{So, } P(10\text{L}) = (2.22)(0.082057)(573.15)$$

$$P = 10.442\text{atm or } 10.6\text{ bar}$$